

Qatar-1b

R-1990

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_200917 · created 2026-08-07T15:00:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459109.6527 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.056
Transit depth (Rp/Rs)^2	2.48 +/- 1.76 [%]
Orbital Inclination (inc)	79.31 +/- 2.79
Airmass coefficient 1 (a1)	51.02 +/- 31.58
Airmass coefficient 2 (a2)	-3.0 +/- 1.65
Scatter in the residuals of the lightcurve fit is	58.87 %
Best Comparison Star	#1 - [76, 68]
Optimal Aperture	3.78
Optimal Annulus	7.82
Transit Duration (day)	0.023 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.158 $\pm$ 0.056 vs 0.1463 (deviation 0.15 σ)
Duration fit vs published	0.023 d vs 0.0692 d (deviation 1.24 σ)
Mid-time O–C	0.1 min
Depth SNR	2.0
Residual scatter	58.87 %

Light curve

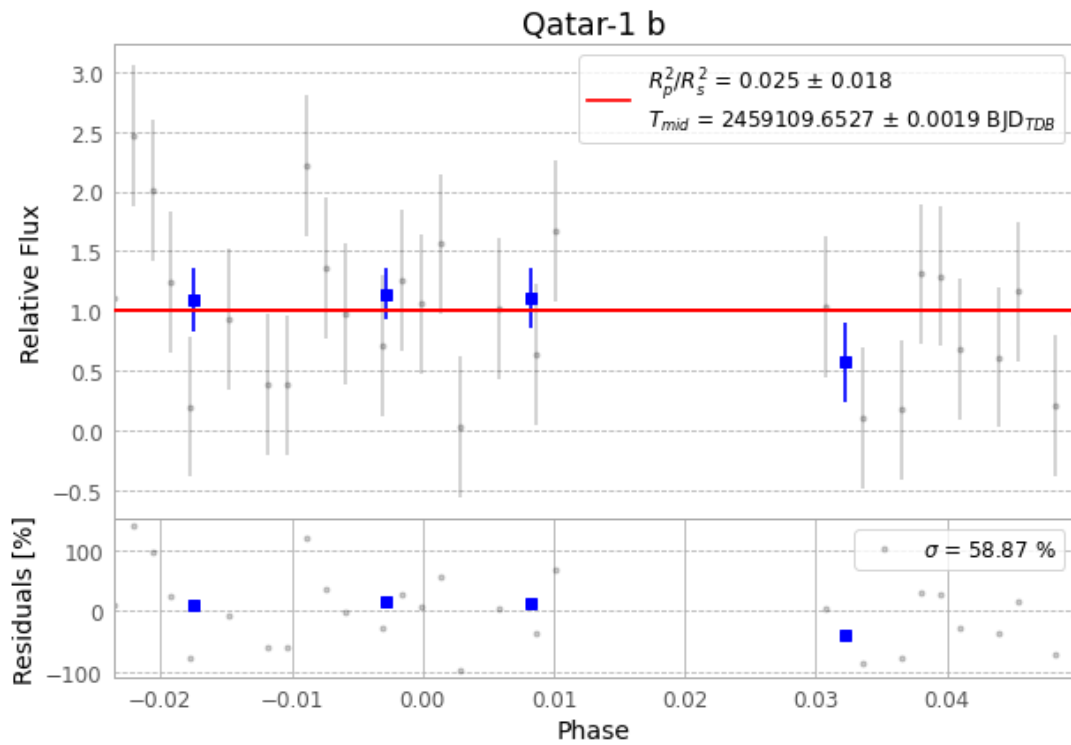


Figure 1 — FinalLightCurve_Qatar-1 b_2020-09-16.png (SHA-256 8d5b8487d80c04a0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [126, 164], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fee7b7a37e35e714...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

54 FITS frames (35 MB), Qatar-1200917024812.FITS → Qatar-1200917052712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-200917.log (ef4159c51579...), FinalLightCurve_Qatar-1 b_2020-09-16.png (8d5b8487d80c...), FinalLightCurve_Qatar-1 b_2020-09-16.csv (97abf02abc7e...)

Compute resources

Wall time 106.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:01Z.